

xNEV: Redefining NHL Offensive Intelligence

A Data-Driven Framework for Evaluating
Team Offensive Systems

OUSAC Final Presentation | Hockey Analytics Project

SYSTEMIC VALUE

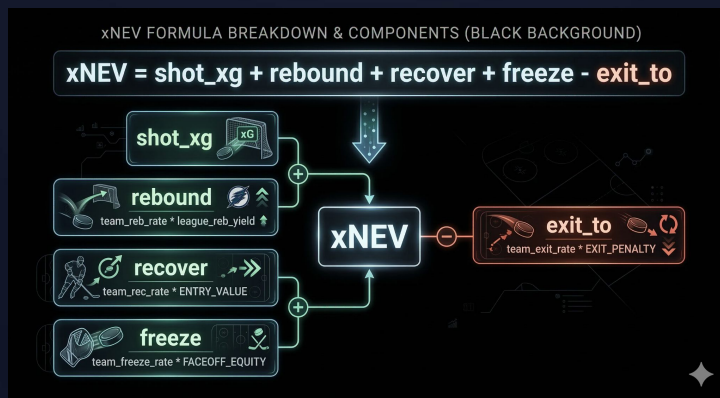
Quantifying the chain of events from
entry to exit.



XNEV FORMULA BREAKDOWN

Primary 5 on 5 shots

$$\text{xNEV} = \text{shot_xg} + \text{rebound} + \text{recover} + \text{freeze} - \text{exit_to}$$



⚽ xG of shot itself

↩ xG of rebound

🔄 Recovery Constant
0.052 xG

❄ Freeze constant
0.023 xG

➡ Exit Constant
-0.026 xG

TRADITIONAL METRICS MISS THE PICTURE

⚠ Current Blind Spots

- ✗ Doesn't account for post shot outcome
- ✗ Volume Illusion



🎯 The xNEV Solution

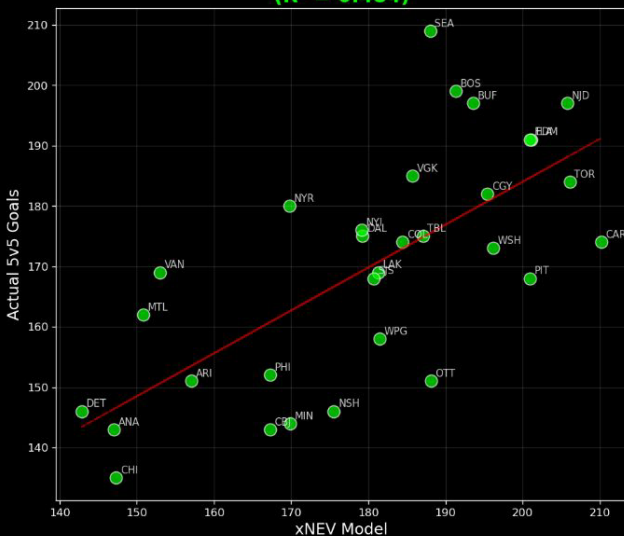
- ✓ Systemic Efficiency
- ✓ Full Sequence Value
- ✓ Predictive Power

VALIDATION & COMPETITIVE BENCHMARKING

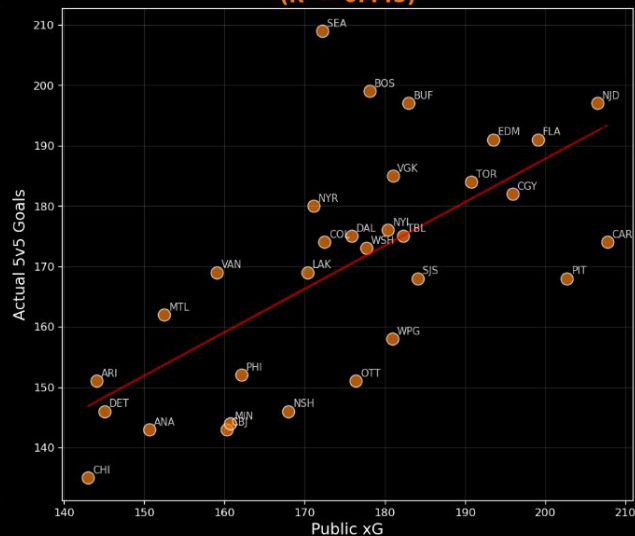
xNEV PERFORMANCE ANALYSIS

The Predictor Race: Which Metric Dominates Goal Prediction?

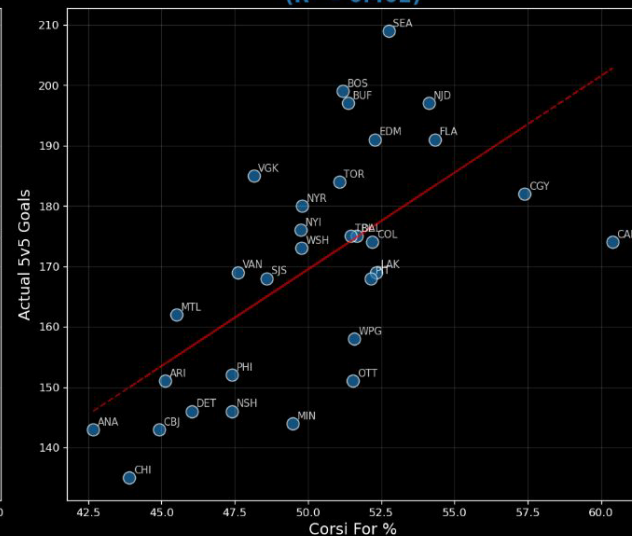
xNEV Model vs Goals
($R^2 = 0.484$)



Public xG vs Goals
($R^2 = 0.443$)

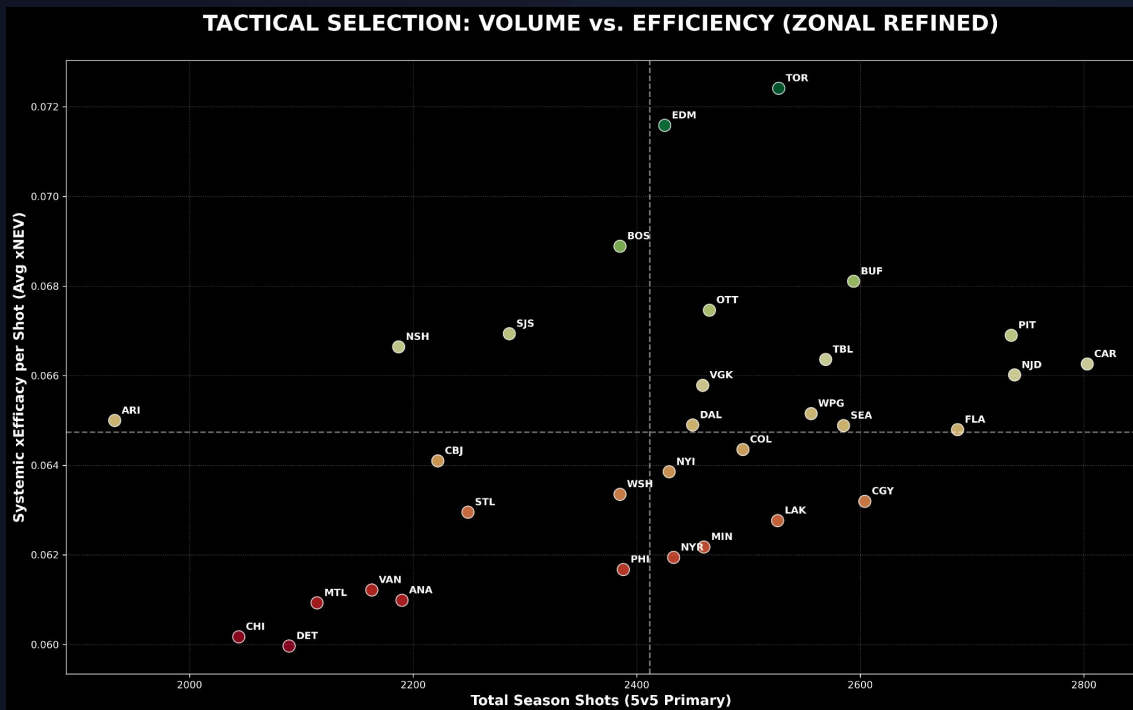


Corsi For % vs Goals
($R^2 = 0.402$)



FOUR DISTINCT OFFENSIVE ARCHETYPES

TACTICAL SELECTION: VOLUME vs.
EFFICIENCY



Efficiency First



TOR, EDM, BOS

Volume First



PIT, NJD, CAR

In Between



BUF, TBL, OTT, VGK

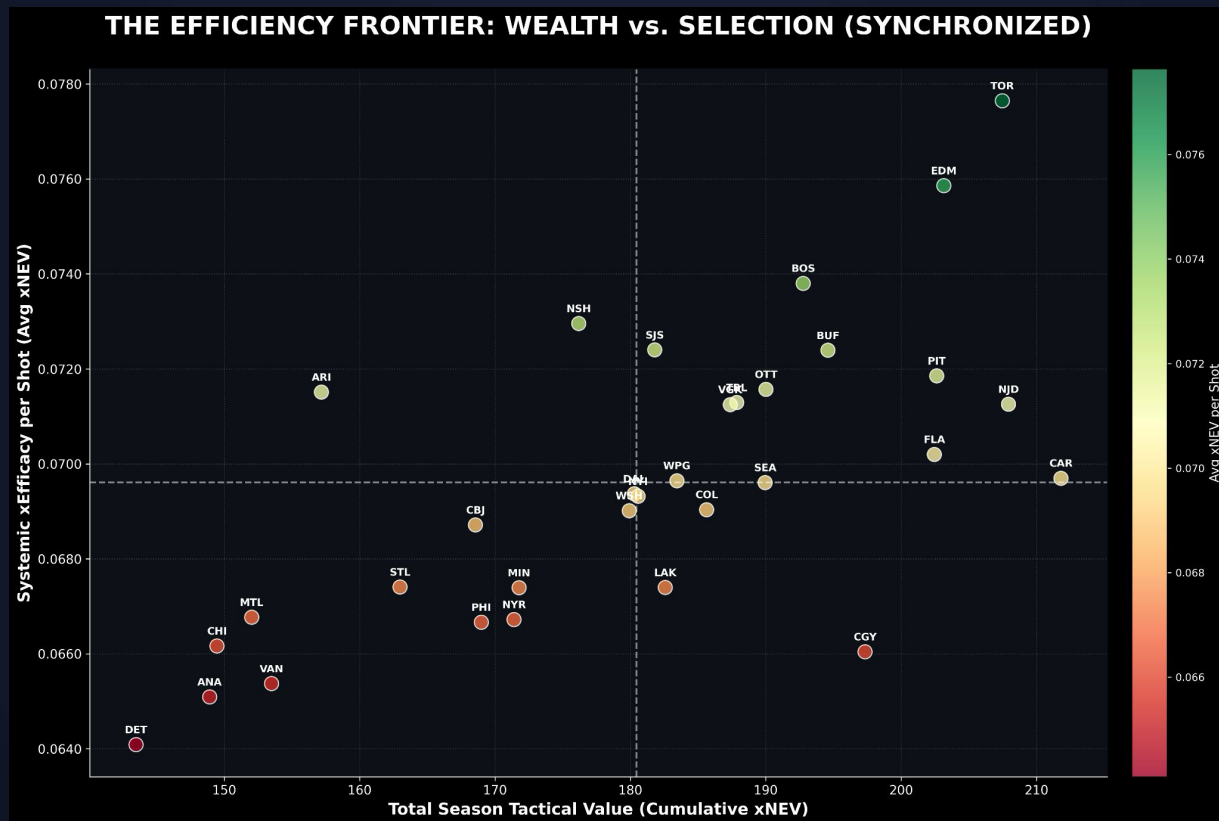
REBUILDING



CHI, DET, ANA

THE EFFICIENCY FRONTIER: WEALTH VS. SELECTION

Multi-Dimensional



ELITE TEAMS DOMINATE THE 12-30 FT ZONE

SYSTEMIC DIFFERENTIATOR RANGE

0-6 FT RANGE

Incredibly Volatile

12-30 FT RANGE

More Realistic and attackable

